X(3872) Analysis Update

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Recap: from reweighting to direct ML selection

Previous workflow

Poor agreement:
MC vs sWeighted data
(sPlot signal)

→ Reweighting (rw) workflow

Current workflow: 2023 PbPb

New NtupleMaker
+ official MC

→ Direct ML selection → fit

Five ML configurations: inputs and selections

Common inputs: `Bmu1pt`, `Bmu1eta`, `Bmu2pt`, `Bmu2eta`, `Btrk2Pt`, `BtrkPtimb`.

Common selection: `Bpt`$<$ 50 GeV, $|$`By`$| < 1.6$, `BQvalue`$<$ 0.15 GeV.

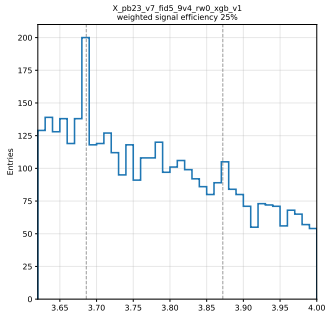
Config.	Additional inputs	`Bpt` (GeV)	`Btrk1dR`	`Btrk2dR`	Sidebands (GeV)
1	`Bmu1phi`, `Bmu2phi`, `Btrk1dR`	> 10	No cut	$(0, 0.25)$	(3.82,3.85); (3.89,3.92)
2	`Bmu1phi`, `Bmu2phi`	> 10	$(0, 0.25)$	$(0, 0.25)$	(3.75,3.85); (3.90,4.00)
3	`Btrk1dR`, `Bujeta`, `Btktketa`	> 10	No cut	$(0, 0.25)$	(3.82,3.85); (3.89,3.92)
4	Same as 3	> 15	No cut	$(0, 0.25)$	(3.82,3.85); (3.89,3.92)
5	Same as 3	> 15	No cut	$(0, 0.35)$	(3.80,3.85); (3.89,3.94)

Sidebands define the training background; intervals have strict bounds.

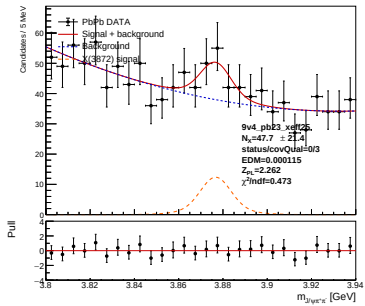
Configuration 1: selected mass spectrum and fit

Relative weighted X-MC efficiency: 25%; score > 0.95279.

cut_scan: selected DATA



Default single-X fit



$N_X = 47.71 \pm 21.44$; $\mu_X = 3876.69 \pm 7.51$ MeV; scale $= 1.500$.

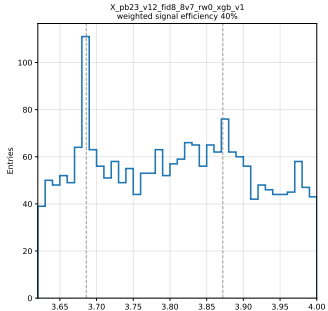
Significance: 2.262.

Mean and scale at their upper bounds. Fit range: 3.80–3.94 GeV.

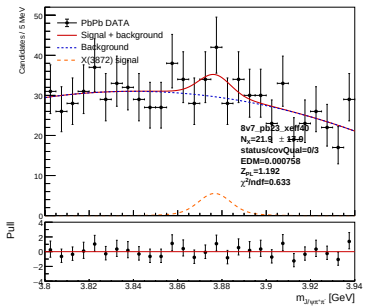
Configuration 2: selected mass spectrum and fit

Relative weighted X-MC efficiency: 40%; score > 0.86855.

cut_scan: selected DATA



Default single-X fit



$N_X = 21.94 \pm 17.92$; $\mu_X = 3876.69 \pm 6.90$ MeV; scale $= 1.500$.

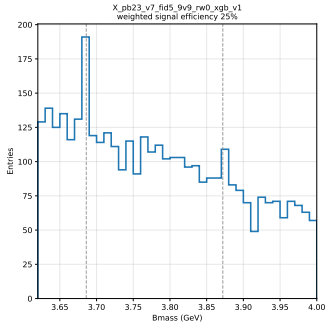
Significance: 1.192.

Mean and scale at their upper bounds. Fit range: 3.80–3.94 GeV.

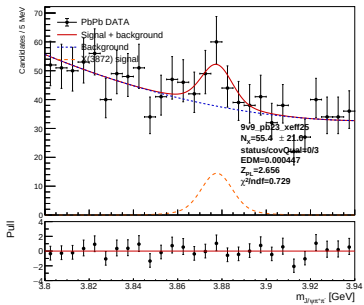
Configuration 3: selected mass spectrum and fit

Relative weighted X-MC efficiency: 25%; score > 0.95267.

cut_scan: selected DATA



Default single-X fit



$N_X = 55.42 \pm 21.04$; $\mu_X = 3877.67 \pm 2.82$ MeV; scale $= 1.495$.

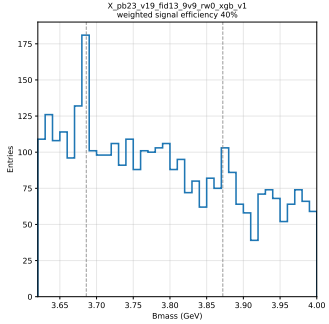
Significance: 2.656.

Mean range: nominal ± 10 MeV; scale range: [0.5,1.5]. Fit range: 3.80–3.94 GeV.

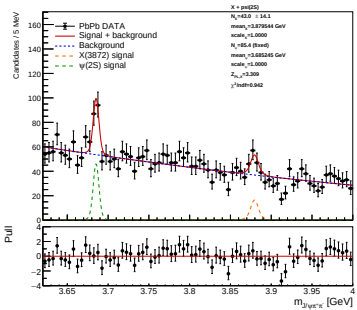
Configuration 4: selected mass spectrum and fit

Relative weighted X-MC efficiency: 40%; score > 0.84505.

cut_scan: selected DATA



Two-peak fit



$N_X = 42.98 \pm 14.11$; $\mu_X = 3879.54$ MeV.

$N_\psi = 85.41$; $\mu_\psi = 3685.24$ MeV.

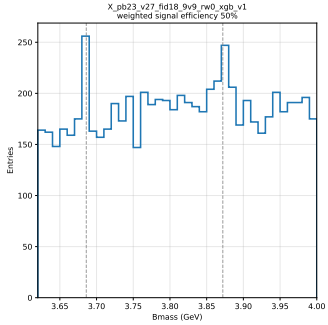
Significance: 3.309.

Means and N_ψ fixed from the first fit; both scales fixed to 1.

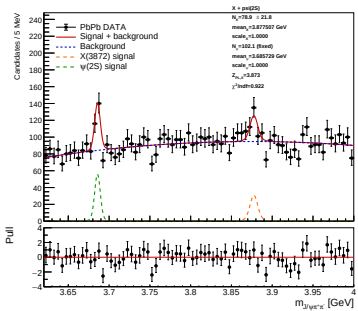
Configuration 5: selected mass spectrum and fit

Relative weighted X-MC efficiency: 50%; score > 0.78272.

cut_scan: selected DATA



Two-peak fit



$N_X = 78.88 \pm 21.76$; $\mu_X = 3877.51$ MeV.

$N_\psi = 102.07$; $\mu_\psi = 3685.73$ MeV.

Significance: 3.873.

Means and N_ψ fixed from the first fit; both scales fixed to 1.

ML recipe comparison: summary

Config.	Main change	ϵ_X	N_X	Significance
1	Nine-input starting point	25%	47.7 ± 21.4	2.262
2	`Btrk1dR`: input $\rightarrow$ selection; wider sidebands	40%	21.9 ± 17.9	1.192
3	Two muon ϕ inputs $\rightarrow$ `Bujeta`, `Btktketa`	25%	55.4 ± 21.0	2.656
4	`Bpt`>15 GeV	40%	43.0 ± 14.1	3.309
5	Wider `Btrk2dR` selection and sidebands	50%	78.9 ± 21.8	3.873

Tested alternatives with worse results:

- The configuration used in the previous analysis.
- Using both `Btrk1dR` and `Btrk2dR` as training inputs.
- Other previously useful variables: `Btktkpt`, `Balpha`, `Bcos_dtheta`, `Bchi2Prob`.

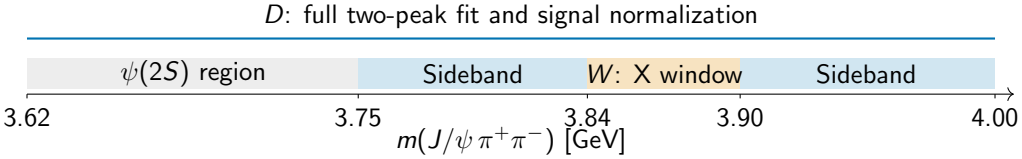
Thanks for listening!

Reporter: Lekai Yao

2026.9.16

Backup: configuration 5 — FOM definition

2023 PbPb; configuration 5 selection. Scan $\epsilon_X = 15$–60% in 1% steps (46 points).



Reference: 50% efficiency, score > 0.782725; use one existing two-peak fit to normalize all cuts. $N_{\mathrm{ref}} = 78.79 \pm 22.14$, $\mu_X = 3877.51$ MeV: first fit with floating means and fixed MC widths.

Expected signal in W

$$S_W(c) = N_{\mathrm{ref}} \frac{\epsilon_D(c)}{\epsilon_D(c_0)} A_W(c).$$

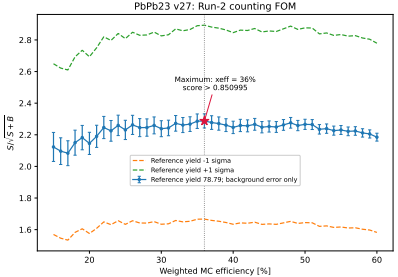
ϵ_D: `pThatreweight`-weighted MC efficiency in D.
A_W: MC double-Gaussian integral ratio W/D; same-cut widths, fixed reference DATA mean and scale 1.

Background in W: unbinned second-order Chebyshev (C2) fit to $(3.75, 3.84) \cup (3.90, 4.00)$, avoiding X and $\psi(2S)$; sideband-count normalization, extrapolated to W.

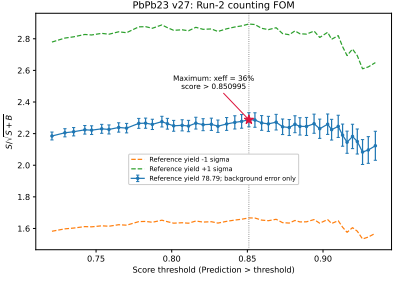
Backup: configuration 5 — FOM scan

$F = S_W/\sqrt{S_W + B_W}$; one fixed 50% reference yield across all 46 cuts.

FOM vs weighted MC efficiency



FOM vs score threshold



Numerical maximum: $\epsilon_X = 36\%$, score > 0.850995; $S_W = 56.66$, $B_W = 556.79 \pm 23.97$.

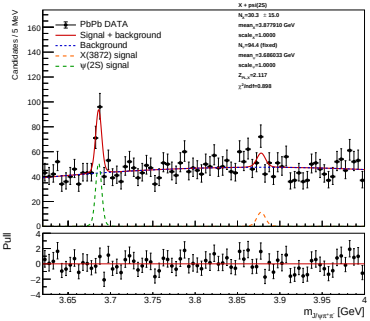
FOM: 2.287799 (36%) vs 2.266093 (50%); improvement 0.958%.

Broad plateau: 35% gives 2.285971; $N_{\mathrm{ref}} \pm 1\sigma$ curves both peak at 36%.

Error bars: background statistics only. Dashed curves: reference-yield variations, not confidence bands. The 36% point was frozen before the new DATA fit; FOM is a counting proxy, not significance.

Backup: configuration 5 — fit at the FOM maximum

Frozen point: $\epsilon_X = 36\%$, score > 0.850995; two-peak fit over 3.62–4.00 GeV.



36% fit parameters

$\mu_X = 3877.91$ MeV (fixed).
$N_\psi = 94.35$, $\mu_\psi = 3686.03$ MeV (fixed).

Means, N_ψ and background shape fixed from the first fit; both MC scales fixed to 1.

	36%	50% reference
FOM	2.288	2.266
N_X	30.33 ± 15.00	78.88 ± 21.76
Significance	2.117	3.873

Higher FOM did not increase the observed significance in this DATA sample.

Broad FOM plateau; no final working point selected.